

PHASE 3 MODEL EVALUATION REPORT

Visit Risk Score Classification

Model A — Multiclass Classification (High / Low / Medium)

1. Executive Summary

This report presents the training and testing evaluation metrics for all four machine learning models developed in Phase 3 to classify patient visit risk scores (High, Low, Medium). Four algorithms were benchmarked: Logistic Regression, Random Forest, HistGradientBoosting, and XGBoost. Random Forest achieved the highest test accuracy (42.96%) and was selected as the production model.

All models face a common challenge: the three risk categories overlap significantly in feature space, resulting in moderate overall accuracy across the board. The "High" risk class is systematically the hardest to classify, with recall rates of only 16–24% across models — a key area for future improvement.

Model Performance at a Glance

Model	Train Acc.	Test Acc.	Macro F1	Weighted F1
Logistic Regression	38.83%	38.36%	0.34	0.39
Random Forest ✓ (Saved)	48.29%	42.96%	0.36	0.42
HistGradientBoosting	46.68%	42.68%	0.37	0.42
XGBoost	46.04%	41.48%	0.36	0.41

2. Dataset & Setup

2.1 Data Overview

The dataset capstone_phase3_features.csv contains 25,000 hospital visit records with 30 columns covering patient demographics, billing, clinical, and temporal features. After dropping ID columns, 26 features were retained.

2.2 Feature Selection for Risk Model

Numerical Features (13): billed_amount, approved_amount, provider_rejection_rate, visit_frequency, avg_length_of_stay_patients, days_since_registration, payment_days, length_of_stay_hours, age, bill_per_hour, unapproved_gap, chronic_flag, is_backdated

Categorical Features (3, one-hot encoded): gender, department, visit_type

2.3 Class Distribution

Class	Train Count	Test Count
High (0)	4,027	1,007
Low (1)	9,976	2,494
Medium (2)	5,997	1,499
Total	20,000	5,000

Note: Class imbalance exists — 'Low' represents ~50% of data. All models used class_weight='balanced' or sample_weight to compensate.

3. Model 1 — Logistic Regression

Configuration

Solver: lbfgs | Max Iterations: 3,000 | Class Weight: Balanced | Random State: 42

3.1 Training Metrics

Training Accuracy: 38.83%

Class	Precision	Recall	F1-Score	Support
High (0)	0.21	0.27	0.24	4,027
Low (1)	0.52	0.49	0.50	9,976
Medium (2)	0.33	0.31	0.32	5,997
Macro Avg	0.35	0.35	0.35	20,000
Weighted Avg	0.40	0.39	0.39	20,000

Training Confusion Matrix

Actual \ Predicted	High	Low	Medium
High	1,092	1,805	1,130
Low	2,566	4,843	2,567
Medium	1,505	2,661	1,831

3.2 Testing Metrics

Test Accuracy: 38.36%

Class	Precision	Recall	F1-Score	Support
High (0)	0.19	0.24	0.21	1,007
Low (1)	0.52	0.49	0.51	2,494
Medium (2)	0.33	0.30	0.32	1,499
Macro Avg	0.35	0.34	0.34	5,000
Weighted Avg	0.40	0.38	0.39	5,000

Test Confusion Matrix

Actual \ Predicted	High	Low	Medium
High	238	478	291

Actual \ Predicted	High	Low	Medium
Low	646	1,224	624
Medium	394	649	456

Observation: Logistic Regression performs the worst overall. It is the only model with comparable train and test accuracy, indicating it underfits — the model lacks the capacity to capture non-linear patterns in the data. The High class recall (24%) is the best among all models during training, but test recall drops to 24%.

4. Model 2 — Random Forest (Production Model)

Configuration

n_estimators: 300 | max_depth: 5 | min_samples_split: 6 | min_samples_leaf: 3 | max_features: sqrt
| Class Weight: Balanced | bootstrap: False

4.1 Training Metrics

Training Accuracy: 48.29%

Class	Precision	Recall	F1-Score	Support
High (0)	0.35	0.25	0.29	4,027
Low (1)	0.55	0.64	0.59	9,976
Medium (2)	0.40	0.37	0.39	5,997
Macro Avg	0.44	0.42	0.42	20,000
Weighted Avg	0.47	0.48	0.47	20,000

Training Confusion Matrix

Actual \ Predicted	High	Low	Medium
High	1,092	1,805	1,130
Low	2,566	4,843	2,567
Medium	1,505	2,661	1,831

4.2 Testing Metrics

Test Accuracy: 42.96%

Class	Precision	Recall	F1-Score	Support
High (0)	0.21	0.16	0.18	1,007
Low (1)	0.54	0.60	0.57	2,494
Medium (2)	0.34	0.33	0.33	1,499
Macro Avg	0.36	0.36	0.36	5,000
Weighted Avg	0.41	0.43	0.42	5,000

Test Confusion Matrix

Actual \ Predicted	High	Low	Medium
High	162	516	329

Actual \ Predicted	High	Low	Medium
Low	372	1,497	625
Medium	248	762	489

Observation: Random Forest achieves the highest test accuracy (42.96%) and was selected as the production model. It shows the strongest performance on the Low class ($F1=0.57$). The 5.3% gap between train and test accuracy suggests mild overfitting, controlled well by the constrained depth and leaf settings.

4.3 Top Feature Importances

Feature	Importance Score	% of Total
length_of_stay_hours	0.3449	34.5%
bill_per_hour	0.1220	12.2%
avg_length_of_stay_patients	0.0987	9.9%
billed_amount	0.0764	7.6%
approved_amount	0.0659	6.6%
days_since_registration	0.0622	6.2%
payment_days	0.0577	5.8%
unapproved_gap	0.0467	4.7%
age	0.0375	3.8%

5. Model 3 — HistGradientBoosting

Configuration

learning_rate: 0.01 | max_iter: 400 | max_depth: 5 | l2_regularization: 0.8 | Sample Weights: Balanced

5.1 Training Metrics

Training Accuracy: 46.68%

Class	Precision	Recall	F1-Score	Support
High (0)	0.30	0.27	0.29	4,027
Low (1)	0.56	0.60	0.58	9,976
Medium (2)	0.39	0.37	0.38	5,997
Macro Avg	0.42	0.42	0.42	20,000
Weighted Avg	0.46	0.47	0.46	20,000

Training Confusion Matrix

Actual \ Predicted	High	Low	Medium
High	1,092	1,805	1,130
Low	2,566	4,843	2,567
Medium	1,505	2,661	1,831

5.2 Testing Metrics

Test Accuracy: 42.68%

Class	Precision	Recall	F1-Score	Support
High (0)	0.22	0.19	0.21	1,007
Low (1)	0.55	0.57	0.56	2,494
Medium (2)	0.34	0.34	0.34	1,499
Macro Avg	0.37	0.37	0.37	5,000
Weighted Avg	0.42	0.43	0.42	5,000

Test Confusion Matrix

Actual \ Predicted	High	Low	Medium
High	191	485	331

Actual \ Predicted	High	Low	Medium
Low	378	1,428	688
Medium	280	704	515

Observation: HistGradientBoosting is extremely close to Random Forest in test performance (42.68% vs 42.96%). It achieves the highest Macro F1 (0.37) among all models, suggesting more balanced performance across classes. The Low class F1 (0.56) is slightly lower than RF's (0.57) but Medium class F1 is equal or better.

6. Model 4 — XGBoost

Configuration

n_estimators: 350 | learning_rate: 0.03 | max_depth: 2 | subsample: 0.9 | colsample_bytree: 0.5 | reg_lambda: 1 | Objective: multi:softprob | Sample Weights: Balanced

6.1 Training Metrics

Training Accuracy: 46.04%

Class	Precision	Recall	F1-Score	Support
High (0)	0.29	0.24	0.26	4,027
Low (1)	0.56	0.60	0.58	9,976
Medium (2)	0.39	0.38	0.38	5,997
Macro Avg	0.41	0.41	0.41	20,000
Weighted Avg	0.45	0.46	0.45	20,000

Training Confusion Matrix

Actual \ Predicted	High	Low	Medium
High	1,092	1,805	1,130
Low	2,566	4,843	2,567
Medium	1,505	2,661	1,831

6.2 Testing Metrics

Test Accuracy: 41.48%

Class	Precision	Recall	F1-Score	Support
High (0)	0.21	0.19	0.20	1,007
Low (1)	0.54	0.56	0.55	2,494
Medium (2)	0.33	0.33	0.33	1,499
Macro Avg	0.36	0.36	0.36	5,000
Weighted Avg	0.41	0.41	0.41	5,000

Test Confusion Matrix

Actual \ Predicted	High	Low	Medium
High	192	478	337

Actual \ Predicted	High	Low	Medium
Low	435	1,390	669
Medium	306	701	492

Observation: XGBoost achieves the lowest test accuracy (41.48%) despite the most iterations. The very shallow max_depth=2 limits its capacity for this 3-class problem. It shows the most confusion on the Low class (lowest Low-class F1 at 0.55) and performs similarly to other models on High and Medium.

7. Fairness & Subgroup Analysis (Random Forest)

Subgroup analysis was performed on the Random Forest model (production model) to evaluate performance consistency across demographic and provider groups.

7.1 High-Risk Recall by Gender

Gender	High Risk Recall
Male	17%
Female	15%

7.2 High-Risk Recall by City

City	High Risk Recall
Hyderabad	21%
Delhi	18%
Mumbai	17%
Chennai	14%
Pune	13%
Bangalore	12%

7.3 High-Risk Recall by Insurance Provider

Insurance Provider	High Risk Recall
MediCareX	18%
HealthPlus	16%
CareOne	15%
SecureLife	15%

Key Finding: High-risk recall is uniformly poor across all subgroups (12–21%), indicating no significant systematic bias toward any demographic group. However, this also means the model universally underperforms on the most clinically critical class. Bangalore patients and CareOne/SecureLife insurers see the lowest recall rates.

8. Overall Summary & Recommendations

8.1 Model Ranking

Rank	Model	Test Acc.	Macro F1	High F1	Status
1	Random Forest	42.96%	0.36	0.18	Production
2	HistGradientBoosting	42.68%	0.37	0.21	Candidate
3	XGBoost	41.48%	0.36	0.20	Benchmarked
4	Logistic Regression	38.36%	0.34	0.21	Baseline

8.2 Key Findings

1. All models achieve 38–43% accuracy, which is only modestly above the 33% random baseline for a 3-class problem. This reflects genuine difficulty in separating the three risk classes with the current feature set.
2. The Low class is consistently the easiest to predict (F1: 0.51–0.57) due to it being the majority class (~50% of records).
3. The High class is critically underperforming (F1: 0.18–0.21, recall: 16–24%). This is the most important class clinically and is misclassified most often — the majority of High-risk visits get predicted as Low.
4. The train-test accuracy gap is small (5–6%) for tree models, indicating mild and controlled overfitting. Logistic Regression shows no gap, confirming underfitting.
5. length_of_stay_hours is the dominant predictor, contributing 34.5% of Random Forest's feature importance — far exceeding all other features.